The Federated Nervous System

The Neuromorphic DHT

A learning-adaptive routing protocol for the AI agent era.

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THE MISSION

AI agents need to find each other.

Without a custodian. At internet scale. In real time.

We built the protocol that does it.

Open-source. Non-profit. Built on twenty-five years of routing research, validated on a 50,000-node simulator, performing within 18% of the analytical lower bound.

THE PROBLEM

~2 seconds

Global lookup latency on the routing protocol every "decentralized AI" project ships today (Kademlia, ~1 M nodes).

Today, every AI interaction routes through 3–5 hyperscalers — OpenAI, Anthropic, Google, Microsoft, Meta. Tomorrow, billions of autonomous agents will need to discover each other, exchange context, and federate work.

The choices today: trust a custodian, or accept multi-second latency that makes real-time impossible.

> ***The AI federation has no nervous system. The substrate doesn't exist.***

WHY NOW

1 billion+

AI agents projected by 2030. Today: zero production-grade routing protocols designed for them.

Five forces converging in 2025–2026:

1. **AI agent proliferation** — autonomous + assisted, billions within five years
2. **$50 B+ already invested in decentralized AI** (Bittensor, Gensyn, Ritual, Akash) — all bottlenecked on Kademlia-class routing
3. **Browser-native P2P** — WebRTC ships everywhere; agents run anywhere
4. **Trust collapse** — AI-generated content + surveillance erode custodial trust at the moment AI scale is exploding
5. **Cryptographic identity is solved** — every agent will have a key. The only missing primitive is *routing*.

> ***Every prerequisite for the AI nervous system exists. The substrate is what's missing.***

THE SOLUTION

100%

Pub/sub baseline delivery. 100% recovered under 5% churn. No replication. No gossip. No parent tracking.

A routing protocol that learns. Hebbian long-term potentiation — the brain's actual learning rule — applied to overlay routing. Every successful path strengthens; unused edges decay. The routing table evolves toward a traffic-shaped optimum.

Two contributions to the AI stack:

It IS an AI model.	Weights, decay, reinforcement, synaptic tagging, plasticity — applied to routing decisions. The first DHT whose routing table *learns from traffic*.
It's the substrate AI federation requires.	Discovery, messaging, pub/sub for billions of agents. Built into the protocol, not layered on a custodial API.

Browser-native. Open-source. Validated in a 50,000-node simulator.

THE PROOF

1.18×

The Dabek 3δ analytical lower bound — within 36 ms of the theoretical optimum at 50,000 nodes.

Dabek et al. (NSDI 2004) proved a hard latency floor for any recursive O(log N) DHT: total ≈ **3δ** where δ is the median pairwise one-way internet latency. The bound has stood unbeaten for twenty years.

Protocol at 50 K nodes	× the floor
Neuromorphic DHT (NX-17)	**1.18×**
Neuromorphic DHT (NH-1, current)	1.28×
Geographic DHT	1.41×
Kademlia *(every existing decentralized AI project)*	**2.65×** *and worsens with scale*

> ***The first DHT measured at the floor. Performance is no longer the obstacle.***

THE MARKET

$1T+

AI infrastructure annual spend projected by 2030. The opportunity isn't capturing share from a platform — it is becoming the protocol underneath.

Layer	Why we win
AI agent infrastructure ($1 T+ by 2030)	We are the missing routing primitive
Decentralized AI compute ($50 B+ today, 50 % YoY)	Bittensor / Gensyn / Ritual all on Kademlia today
Decentralized identity ($10 B+)	Every agent has an identity; needs routing
Privacy-respecting messaging ($5 B+)	Direct substitute for Signal-class infrastructure
Federated learning + edge AI	Currently impossible without a substrate

> ***TCP/IP, HTTP, DNS — none owned. The AI federation's nervous system should be the same.***

THE COMPETITION

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Production-grade routing protocols at the analytical floor. Until now.

Class	Speed	Custodian	Adapts to traffic
Centralized AI APIs (OpenAI, Anthropic)	fast	**YES**	n/a
Kademlia DHTs (IPFS, BitTorrent, Ethereum)	~2 s	none	no
"Decentralized AI" stacks (Bittensor, Gensyn, Ritual)	~2 s *(inherits)*	partial	no
Federated platforms (Mastodon, Matrix)	server-bound	**YES**	no
Neuromorphic DHT	**~250 ms**	**none**	**YES**

The moat: twenty-five years of compounded routing research — Kademlia → Pastry → Tapestry → SCRIBE → NX-17 → NH-1 — performing at the analytical floor, open-source for adoption velocity.

THE ROADMAP

18 months

From research-grade today to production-grade NH-1 in deployment. Milestones defined and falsifiable.

Phase	Duration	Outcome
Now	—	50K-node simulator, IEEE paper, three red-team analyses, all open-source
0–6 months	6 mo	Production hardening — connection-setup model, RPC timeouts, bandwidth (red-team Phase 1 deploy blockers)
6–12 months	6 mo	Three reference applications: agent registry, agent-to-agent messaging, federated-learning coordination
12–18 months	6 mo	SDKs, browser integration, developer ecosystem
18–36 months	18 mo	100K+ nodes in production; first commercial pilots in regulated industries

> ***Phase 1 deploy-blocker work is scoped at 6–8 weeks for two engineers.*** *The path to production is short, scoped, and starts with falsifiable milestones.*

THE VISION

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Substrate layers, none owned by any single party.

Layer	Year	What it enabled
TCP/IP	1974	Internet routing
DNS	1983	Name resolution
HTTP	1990	The web
Linux	1991	Open-source operating systems

The 5th is the AI federation's nervous system. That's what we're building.

> ***The alternative is the same five hyperscalers that own AI compute owning AI routing.*** *That outcome is the one this project exists to prevent. Privacy and intelligence — both built into the routing fabric, not bolted on after.*

THE ASK

$10M

24 months. Becomes infrastructure no one owns.

Allocation	%	What it buys
Production engineering	60 %	2–3 senior protocol engineers, scaled simulator, deploy-blocker mitigation, 1 M-node validation
Reference applications	25 %	Three high-profile partner deployments demonstrating what becomes possible
Developer ecosystem	15 %	SDKs, documentation, conference presence, ecosystem partnerships

Open-source. Non-profit. Funding becomes a public good — like TCP/IP, HTTP, DNS, and Linux were before private capital colonized the layers above. The AI federation deserves the same architectural commitment.

> ***Without a public-good substrate, the same five hyperscalers will own routing too. Fund the alternative now, while it's still possible.***